

Machine Learning Models for Prediction

Siyabonga Mavundla
South African Medical Research Council (SAMRC)

African Data Science Conference, Johannesburg
June 2026

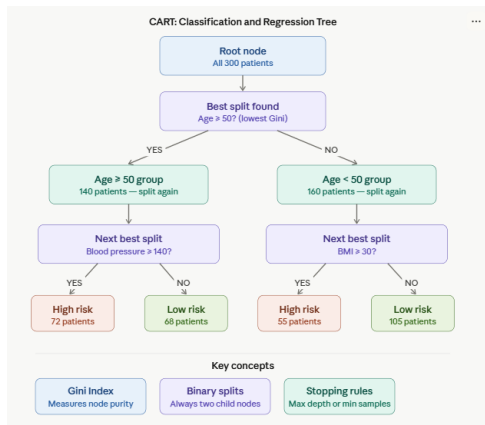
Overview

- ▶ We explore common machine learning models used for prediction
- ▶ Focus on:
 - ▶ Tree-based models: CART, Random Forest, Gradient Boosting
 - ▶ Survival models: Random Survival Forests
 - ▶ Other methods: SVM and k-NN
- ▶ Goal: understand how they work and when to use them

How to Think About These Models

- ▶ CART makes decisions using simple step-by-step rules
- ▶ Random Forest combines many decision trees to improve stability
- ▶ Gradient Boosting improves predictions by learning from mistakes
- ▶ RSF predicts outcomes over time (e.g. patient survival)
- ▶ SVM separates groups using an optimal boundary
- ▶ k-NN groups patients based on similarity

CART: Intuition



- ▶ CART works like a decision flowchart
- ▶ At each step, the data is split based on a simple rule
- ▶ For example, we might ask:
 - ▶ Is age less than 50?
 - ▶ Is blood pressure high?
- ▶ These splits help group similar patients together

CART: Key Ideas

- ▶ CART is a supervised learning model used for prediction
- ▶ It splits data into smaller groups based on features
- ▶ The goal is to create groups that are as uniform as possible
- ▶ This is measured using metrics like Gini Index or Entropy
- ▶ One limitation is that deep trees can overfit the data

CART: Advantages & Disadvantages

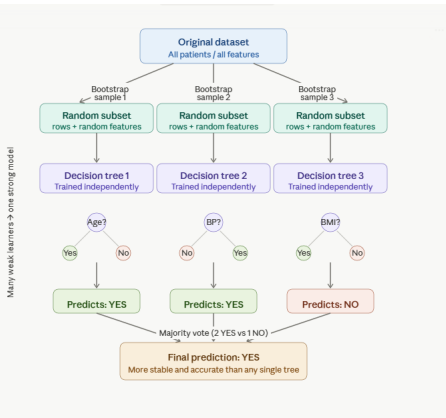
Advantages

- ▶ Highly interpretable
- ▶ Handles mixed data types (Numerical / Categorical)
- ▶ No need to rescale or normalize your data

Disadvantages

- ▶ High variance (unstable)
- ▶ Overfitting without pruning

Random Forest: Intuition



- ▶ Random Forest builds many decision trees instead of one
- ▶ Each tree sees slightly different data
- ▶ Each tree makes its own prediction
- ▶ The final result is based on majority voting

Random Forest: Key Ideas

- ▶ Random Forest is an ensemble learning method
- ▶ It reduces overfitting by averaging many trees
- ▶ It uses:
 - ▶ Random sampling of data (bootstrapping)
 - ▶ Random selection of features
- ▶ This makes the model more stable and reliable

Random Forest: Advantages & Disadvantages

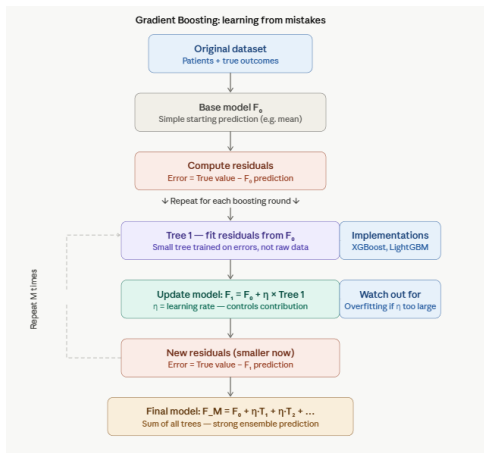
Advantages

- ▶ **High Predictive Accuracy:** Combines predictions from many uncorrelated decision trees, outperforming a single decision tree
- ▶ **Low Risk of Overfitting:** Averaging multiple trees reduces variance and stabilizes the model
- ▶ **No Feature Scaling Required:** Works directly with raw, unscaled data

Disadvantages

- ▶ **High Computational Cost:** Requires significant memory and processing power to train and store many deep trees

Gradient Boosting: Intuition



- ▶ Gradient Boosting builds models one after another
- ▶ Each new model focuses on correcting previous errors
- ▶ Over time, the predictions become more accurate
- ▶ It is like learning from mistakes step-by-step

Gradient Boosting: Key Ideas

- ▶ Models are added sequentially (not independently)
- ▶ Each step minimizes prediction error
- ▶ A learning rate controls how much each tree contributes
- ▶ It often achieves high accuracy but requires careful tuning

Gradient Boosting: Advantages & Disadvantages

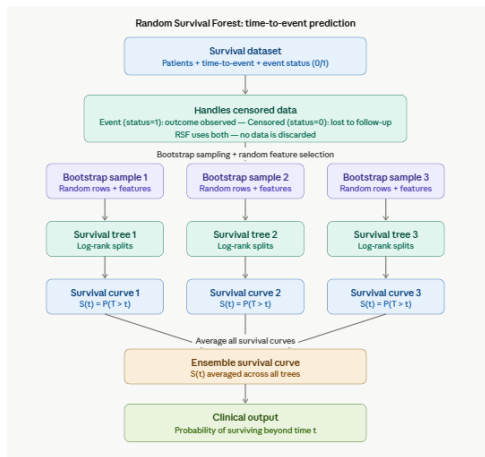
Advantages

- ▶ **Exceptional Predictive Accuracy:** Often achieves state-of-the-art performance on tabular datasets
- ▶ **Handles Complex Patterns:** Captures non-linear relationships and feature interactions
- ▶ **Handles Imbalanced Data:** Focuses on difficult-to-predict instances via residual learning

Disadvantages

- ▶ **High Risk of Overfitting:** Can memorize noise if not properly regularised
- ▶ **Hyperparameter Tuning Required:** Performance depends on learning rate, depth, and number of trees
- ▶ **Slow Training:** Sequential nature prevents parallel tree building

Random Survival Forest: Intuition



- ▶ RSF is used when the outcome is time-related
- ▶ For example: time until recovery or death
- ▶ It combines multiple trees to estimate survival probability
- ▶ Output is a survival curve over time

RSF: Key Ideas

- ▶ Extension of Random Forest for survival data
- ▶ Can handle censored data (incomplete observations)
- ▶ Useful in clinical and epidemiological studies
- ▶ Performance is often measured using the C-index

RSF: Advantages & Disadvantages

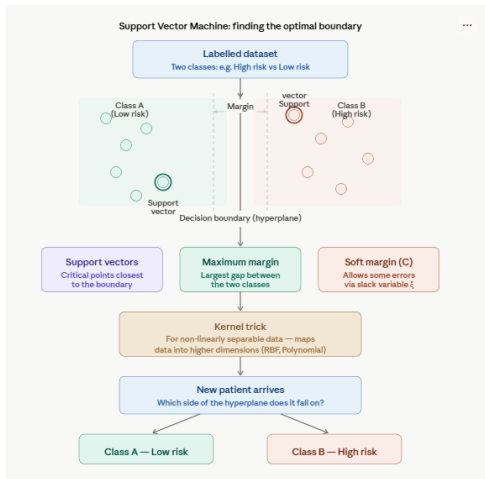
Advantages

- ▶ **No Strict Assumptions:** Does not require the proportional hazards assumption
- ▶ **Handles Complex Interactions:** Captures non-linear relationships automatically
- ▶ **Built-in Validation:** Uses out-of-bag data for performance metrics (e.g., C-index)

Disadvantages

- ▶ **Computationally Intensive:** Requires high memory and processing power
- ▶ **Requires Large Datasets:** Performs poorly on small datasets
- ▶ **Bias Toward Continuous Variables:** Favours variables with many split points

SVM: Intuition



- ▶ SVM separates groups using a boundary (line or plane)
- ▶ The goal is to create the largest possible gap between groups
- ▶ Only a few critical points determine the boundary

SVM: Key Ideas

- ▶ SVM finds an optimal separating hyperplane
- ▶ Uses kernels to handle complex relationships
- ▶ Works well in high-dimensional data
- ▶ Sensitive to noise and overlapping classes

SVM: Advantages & Disadvantages

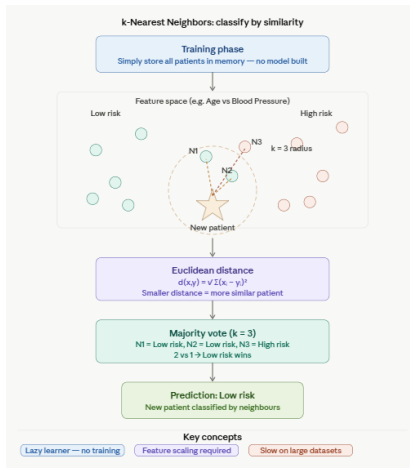
Advantages

- ▶ Highly effective in **high-dimensional spaces**
- ▶ Memory efficient (uses support vectors only)
- ▶ Flexible through kernel functions

Disadvantages

- ▶ Not suitable for large datasets
- ▶ Sensitive to noise and overlapping classes

k-NN: Intuition



- ▶ k-NN classifies a new point based on its neighbours
- ▶ It looks at the most similar patients in the dataset
- ▶ The majority class determines the prediction

k-NN: Key Ideas

- ▶ No explicit training phase (memory-based)
- ▶ Depends on distance between data points
- ▶ Requires feature scaling for good performance
- ▶ Can be slow for large datasets

k-NN: Advantages & Disadvantages

Advantages

- ▶ Simple and intuitive
- ▶ No training phase required
- ▶ Easily adapts to new data

Disadvantages

- ▶ Computationally expensive for large datasets
- ▶ High memory requirement
- ▶ Sensitive to feature scaling and noise

Example: k-NN in Healthcare

- ▶ k-NN has been used to classify diabetes severity
- ▶ Patients are compared based on clinical measurements
- ▶ Patients with similar profiles are grouped together
- ▶ This helps identify risk levels and guide treatment decisions

Model Comparison

Model	Strength	Weakness
CART	Interpretable	Prone to overfitting
Random Forest	Robust, reduces variance	Computationally expensive
Gradient Boosting	High predictive accuracy	Requires tuning
RSF	Suitable for survival data	Computationally intensive
SVM	Effective in high-dimensional spaces	Sensitive to noise
k-NN	Simple and intuitive	Slow for large datasets

R Packages for Machine Learning

- ▶ **CART:**

- ▶ `rpart`, `caret`

- ▶ **Random Forest:**

- ▶ `randomForest`, `ranger`

- ▶ **Gradient Boosting:**

- ▶ `xgboost`, `gbm`, `lightgbm`

- ▶ **Random Survival Forests (RSF):**

- ▶ `randomForestSRC`, `survival`

- ▶ **SVM:**

- ▶ `e1071`, `kernlab`

- ▶ **k-NN:**

- ▶ `class`, `caret`

Selected Applications in Healthcare (I)

▶ CART

- ▶ Olaniyi et al. (2022). *A classification and regression tree algorithm for heart disease modeling and prediction*. ScienceDirect.
- ▶ Fontanet et al. (2020). *Proposed clinical indicators for efficient screening and testing for COVID-19 from CART analysis*. medRxiv.

▶ Random Forest

- ▶ Boppana et al. (2025). *DiabCompSepsAI: Integrated AI model for early detection of postoperative complications in diabetic patients*. Journal of Clinical Medicine.
- ▶ Orji & Ukwandu (2024). *Random forest models in prognostication of gastrointestinal malignancies: a systematic review*. Frontiers.

▶ Gradient Boosting

- ▶ Wang et al. (2022). *Clinical prediction of heart failure in hemodialysis patients using XGBoost*. Frontiers in Genetics.
- ▶ Kuo et al. (2023). *XGBoost-based simple three-item model for predicting outcomes of acute ischemic stroke*. PMC.

▶ Random Survival Forest

- ▶ Zhao et al. (2024). *RSF model for survival prediction in breast cancer patients post-chemotherapy*. JMIR.
- ▶ Scientific Reports (2024). *RSF algorithm for risk stratification in gastric neuroendocrine neoplasms*.

Selected Applications in Healthcare (II)

► SVM

- Qaiser et al. (2024). *SVM outperforms other models in early diagnosis of dengue using routine clinical data*. PMC.
- Hemrungronj et al. (2021). *Accuracy of SVM for diagnosis of Alzheimer's disease using brain volume data*. Frontiers in Neurology.

► k-NN

- Uddin et al. (2022). *Comparative performance analysis of KNN variants for disease prediction*. Scientific Reports.
- BMC (2024). *Automated approach to predict diabetic patients using KNN imputation*. BMC Medical Research Methodology.

Conclusion

- ▶ Different models work better for different problems
- ▶ Tree-based methods are strong for clinical data
- ▶ Survival models are important for time-to-event outcomes
- ▶ The best model depends on the data and the question

Thank You

THANK YOU!